Demon Slayer Corps - The String Hunt

Mission 1 - streaming KMP (4096-character chunks)

```
+ python mission1.py
+ tee ../output/mission1.txt
pi = [0, 0, 1, 2, 3, 4]
Alarm count: 415
First 10 shifts: [8620, 8622, 8624, 39118, 39126, 88124, 88132, 90213, 90215, 90217]
```

Mission 2 - finite automaton and both DNA strands

```
+ python mission2.py
+ tee ../output/mission2.txt
Marker: GATTACAGATTC  Reverse complement: GAATCTGTAATC
delta: state  A  C  G  T
           0  0  0  1  0
           1  2  0  1  0
           2  0  0  1  3
           3  0  0  1  4
           4  5  0  1  0
           5  0  6  1  0
           6  7  0  1  0
           7  0  0  8  0
           8  9  0  1  0
           9  0  0  1 10
          10  0  0  1 11
          11  5 12  1  0
          12  0  0  1  0
Shifts refer to the stored forward sequence for both strands.
sample_01.txt: forward=0 []; reverse=0 []
sample_02.txt: forward=0 []; reverse=0 []
sample_03.txt: forward=0 []; reverse=0 []
sample_04.txt: forward=0 []; reverse=0 []
sample_05.txt: forward=0 []; reverse=0 []
sample_06.txt: forward=0 []; reverse=0 []
sample_07.txt: forward=0 []; reverse=3 [125858, 407774, 668245]
sample_08.txt: forward=0 []; reverse=0 []
sample_09.txt: forward=0 []; reverse=0 []
sample_10.txt: forward=0 []; reverse=0 []
Demon sample(s): sample_07.txt
```

Final Selection: 24/24 PASS. Additional regression tests: PASS.

All matches use zero-based shifts. Overlapping occurrences count.

Sample 07 has three reverse-strand hits and no forward-strand hits.

Screenshots show actual terminal execution of the submitted scripts.

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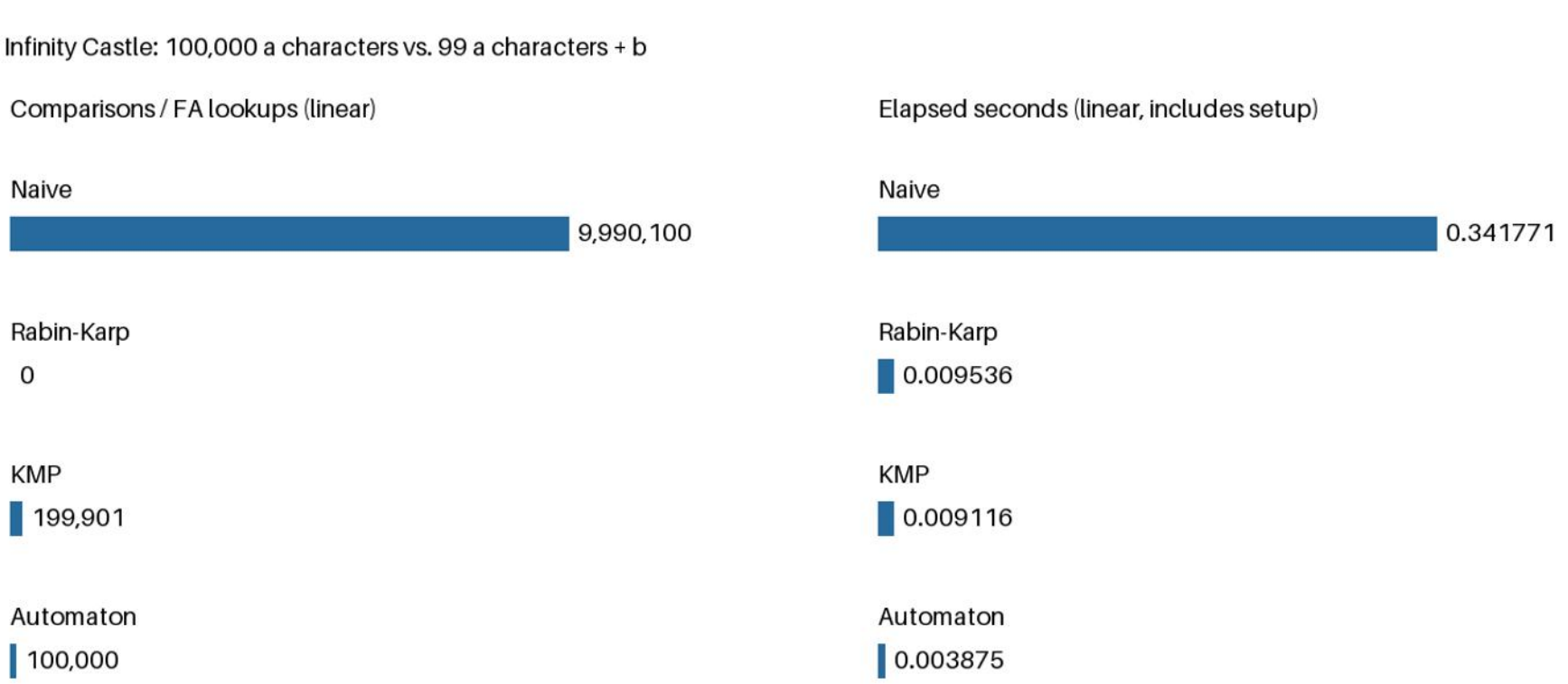
Mission 3 - verified 40-character rolling windows

```
+ python mission3.py
+ tee ../output/mission3.txt
1. Zenitsu   89.00%
2. Kanao     31.86%
3. Inosuke   14.96%
4. Genya      0.00%
5. Tanjiro    0.00%
Changing words breaks exact 40-character windows, leaving fewer verified matches.
The score measures verbatim coverage, so it underestimates paraphrased copying.
```

Inosuke paraphrased passages; his exact-overlap score is only 14.96%.

Bonus - Infinity Castle tournament

```
+ python mission4.py
+ tee ../output/mission4.txt
Times include preprocessing; RK uses d=256, q=101.
Character comparisons; automaton reports table lookups instead.
Automaton       100,000 operations  0.003875 s
KMP             199,901 operations  0.009116 s
Rabin-Karp            0 operations  0.009536 s
Naive         9,990,100 operations  0.341771 s
Automaton wins this run; FA scans with one table lookup per character.
Naive loses: all 100 characters are compared at each of 99,901 shifts.
KMP reuses prefixes; FA uses one lookup per character; RK rejects hashes without verification.
```



Timing includes preprocessing. Counts exclude preprocessing.

FA counts table lookups; RK counts only verification comparisons.

RK still computes rolling hashes even when verification count is zero.